Software Requirements Specification

for

MEETSPACE Smart Conference Room

Version 1.2

28 October 2026

Submitted by

Phakhin Nantakarn 6831503064

Ratchata Suppattapan 6831503067

Witthawin Tanaset 6831503070

Siwakorn Jaronsak 6831503072

Thanaphathr Phutthiphiriya 6831503116

School of Applied Digital Technology, Mae Fah Luang University

Course: Software Requirements Analysis and Specification

Table of Contents

Table of Contents ..... **ii**
Revision History ..... **iii**
Document Approval ..... **iii**
1. Introduction **1**
1.1 Business Problem 1
1.2 Business Objective 1
1.3 Product Vision 1
1.4 Business Scope 1
1.5 Business Workflow 2
1.6 Definitions, Acronyms and Abbreviations 3
2. Overall Description **4**
2.1 Product Perspective 4
2.2 User Classes and Characteristics 6
3. Software Specific Requirements **7**
3.1 Overall Functional Requirements 7
3.2 Functional Requirements 8
3.3 Non-Functional Requirements 15
4. User Interface Requirements **17**
4.1 Navigation Structure / Sitemap 17
4.2 User Interface Specifications 18
Appendix A: Data Dictionary **32**

Revision History

No.	Name	Date	Description	Version
1	MEETSPACE project team	23/10/2026	Initial document based on the project presentation, analysis diagrams and application source.	1.0
2	MEETSPACE project team	27/10/2026	Aligned FR-06, FR-09, FR-10, FR-12, sections 1, 4.1, 4.2 and Appendix A with the source.	1.1
3	Phakhin Nantakarn	28/10/2026	Restructured to the MFU SRS template (FR/NFR/UI tables). Re-verified every requirement against the source code; redrew workflow, use case, sitemap and ER diagrams; replaced UI screenshots; added UI-06 to UI-13, NFR-10 to NFR-13, input limits and error messages.	1.2

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Name	Role	Signature	Date
To be completed	Project Sponsor / Stakeholder		
To be completed	Project Manager (team representative)		

1. Introduction

This document specifies Release 1 of MEETSPACE, a web-based meeting room reservation system with a member website and a separate walk-in kiosk. Every requirement was checked against the application source code (Spring Boot ApiController, schema.sql and the browser application app.js) on 28 October 2026. Features that exist only as legacy fields in the code are listed as out of scope.

1.1 Business Problem

Staff who need a meeting room do not have complete information about each room's equipment and capacity, and the room status they see is not reliable: a room may look free while someone is already using it. Informal bookings (chat messages, verbal agreements) are kept in different places, so double bookings happen, staff lose time searching for a room for urgent or important meetings, and nobody can tell who is responsible for a reservation.

These problems affect employees who book rooms, people who arrive at a room without an account and want to use it immediately (walk-in), and the administrators who manage the rooms.

1.2 Business Objective

MEETSPACE aims to:

- Keep one shared reservation record for both advance (member) and walk-in bookings, so that no meeting room time slot is reserved twice.
- Show room facilities, capacity, photos and occupied time slots before a user books, and confirm a valid request immediately.
- Allow administrators to identify the booker and channel of every reservation and to cancel reservations when necessary.

1.3 Product Vision

For employees who need meeting rooms and the administrators who manage them, MEETSPACE will provide room discovery, shared real-time availability and immediately confirmed reservations through a member website and a separate walk-in kiosk, so that staff can find a suitable room quickly and conflicting bookings are prevented.

1.4 Business Scope

Release 1 includes:

- Public browsing of rooms, facilities, photos and occupied time slots (08:00-20:00) without signing in.
- Account registration, sign in, sign out and change password.
- Member reservations that are confirmed immediately (one room, one date, start and end time).
- Walk-in reservations from a separate kiosk page (/#walkin) using a declared name or employee ID, without an account.
- Server-side validation of every reservation (opening hours, 30-minute slots, maximum 4 hours, not in the past, at most one year ahead, no overlap, active room, no duplicate submission).
- My reservations, reservation history, cancellation and in-app notifications.
- Administration: booking management (date and room filters, separate walk-in and member lists), room management (add, edit, photo, soft delete) and members and roles.
- English and Thai interface, light and dark theme.

Release 1 does not include:

- Approval workflow (pending / approve / reject). Legacy fields and a server action remain in the code but are not reachable from the user interface.
- Recurring (weekly) reservations and participant count. The server fixes repeat = 1 and participants = 1.
- Editing an existing reservation. A user cancels and creates a new reservation instead.
- Email, SMS or Google Calendar notifications (only in-app notifications).
- Verification of a walk-in booker's identity against an employee directory.
- Physical occupancy sensors, payments and deleting user accounts.

1.5 Business Workflow

Member workflow: Sign in; browse rooms and inspect facilities and the availability timeline; choose a room, date, start and end time; review and submit. The server validates the request; if it is valid and the slot is free, the reservation is saved as Confirmed together with a confirmation event. The member sees the result in My reservations and Notifications and may cancel it before its end time. An invalid or conflicting request shows an error message and nothing is saved.

Walk-in booker workflow: Open the kiosk page (/#walkin) without signing in; choose a room, date and time; enter a name or employee ID; review and submit. The server applies the same validation and availability rules and records the reservation under the system walk-in account with the declared booker identity. The kiosk shows a booking reference.

Administrator workflow: Sign in; open Administration > Booking management; filter by date and room; inspect the booker and channel (walk-in or member) and cancel an active reservation if required. Room management and members and roles are also restricted to administrators.

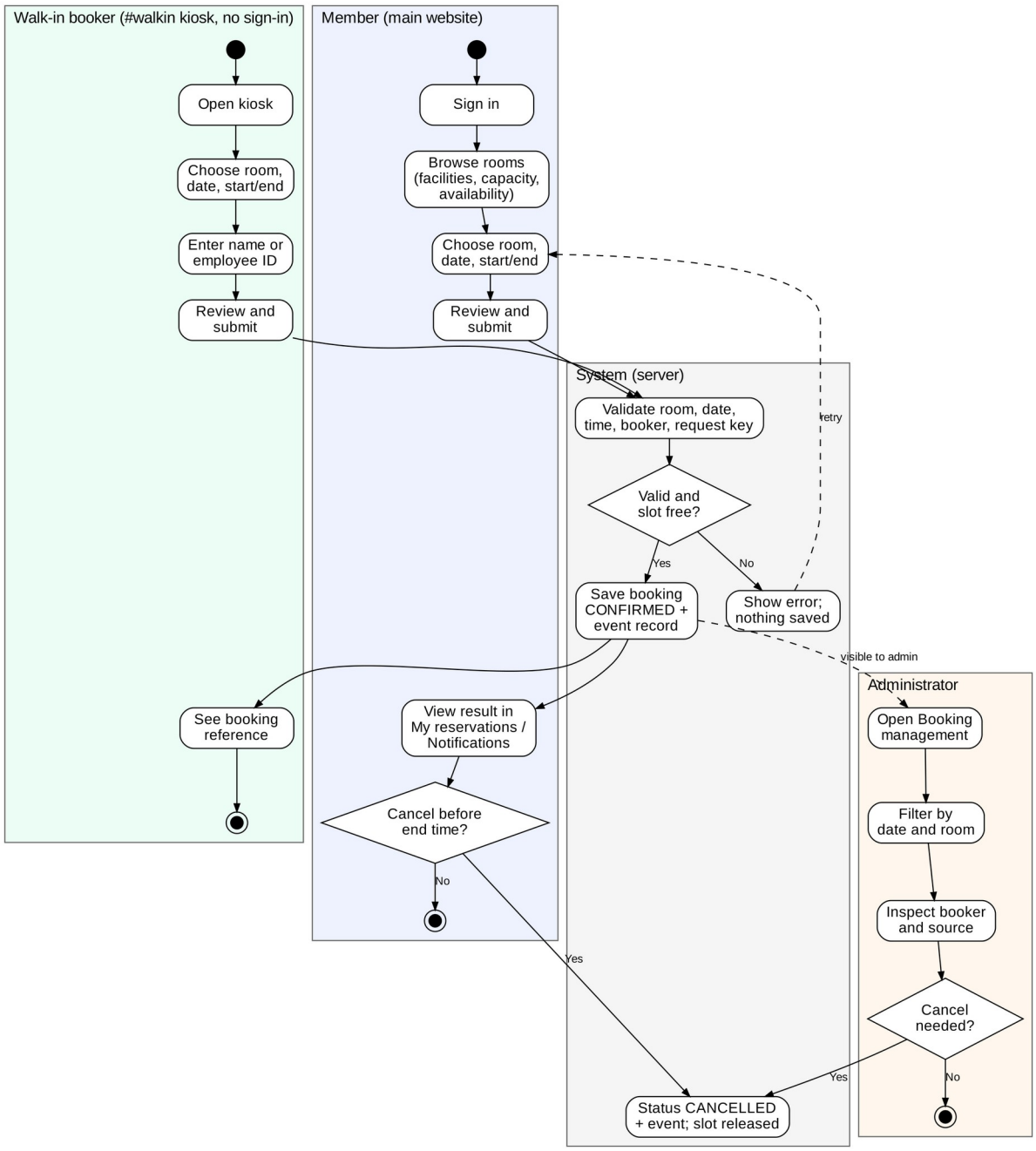


Figure 1 Business workflow (activity diagram with swimlanes)

1.6 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
BR / FR / NFR	Business requirement / functional requirement / non-functional requirement
UI	User interface
DB	Database

Term	Description
PK / FK / UK	Primary key / foreign key / unique key
API	Application programming interface; the REST endpoints under /api
H2	Embedded relational database used to store MEETSPACE data in a file
BCrypt	One-way password hashing function used to store passwords
Visitor	A person who uses the main website without signing in
Member	A signed-in user with role user
Administrator	A signed-in user with role admin
Walk-in	Reservation channel on the separate kiosk page /#walkin that does not require an account
Booker	The name or employee ID typed by a walk-in user, or the member's account name
Confirmed	Reservation status that holds a room time slot
Cancelled	Reservation status after the owner or an administrator cancels it; the time slot is released
Pending / Rejected	Legacy approval statuses kept in the data model; not produced by the Release 1 user flow
Active reservation	A reservation whose status is Pending or Confirmed; used for availability and overlap checks
Request key	Random identifier sent with each booking submission to detect a duplicate submission
Soft delete	Marking a room as deleted (deleted = TRUE) so it is hidden but its reservation history is kept

2. Overall Description

2.1 Product Perspective

MEETSPACE is a stand-alone web application. The browser interface (HTML, CSS and JavaScript) calls Spring Boot REST endpoints under /api, and the server stores accounts, rooms, reservations, events and uploaded room photos in a file-based H2 database (data/meetspace.mv.db) that persists after a restart. The main website and the walk-in kiosk are two interfaces over the same data. All date and time validation on the server uses the Asia/Bangkok time zone.

The system is not integrated with an employee directory, email, Google Calendar or room sensors. A walk-in employee ID is recorded as a declaration and is not authenticated.

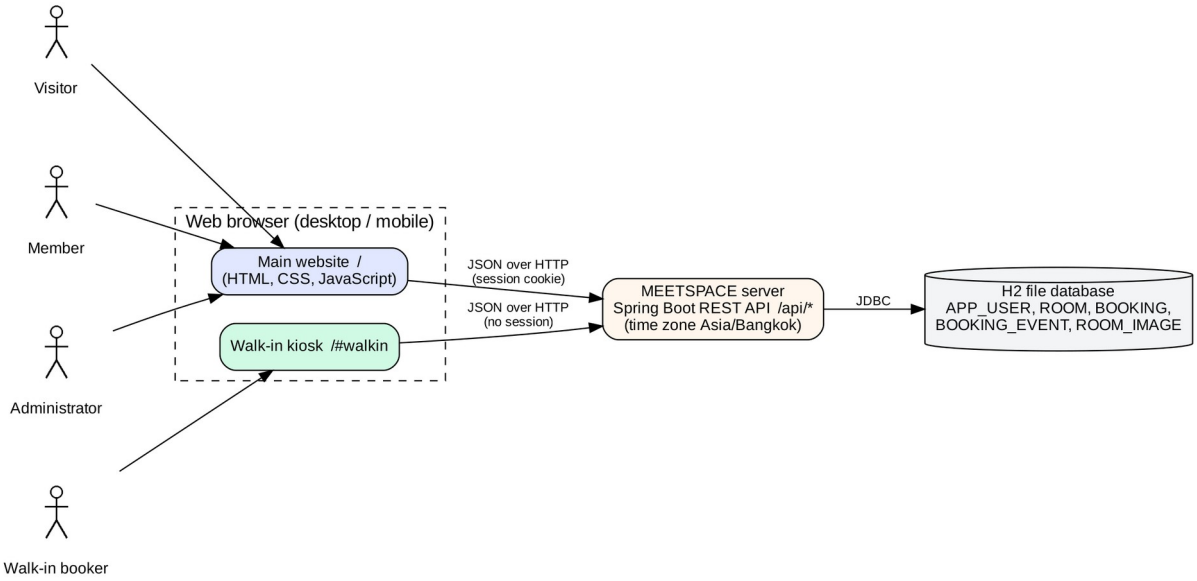


Figure 2 System context

Functions available to each user class:

Visitor

- Browse rooms, facilities and photos
- View anonymous occupied time slots
- Register an account
- Use the walk-in kiosk

Member

- Sign in / sign out
- Create a reservation
- View my reservations, history and notifications
- Cancel own reservation
- Change password

Walk-in booker

- Create a walk-in reservation with a name or employee ID

Administrator

- All member functions
- View and cancel all reservations
- Manage rooms and photos
- Add members and change roles

Constraints and assumptions

- [Constraint] Release 1 runs as a single server instance; reservation requests are serialized inside that instance.

- [Constraint] The prototype seeds a default administrator (admin@meetspace.local / admin123) and enables the H2 web console at /h2-console with an empty database password. Both shall be changed or disabled before production use.
- [Assumption] Users access the system with a current desktop or mobile web browser that has JavaScript enabled.
- [Assumption] Opening hours are 08:00-20:00 every day; the system does not model public holidays.

2.2 User Classes and Characteristics

User Class	Description	Role / Goal	Main Functions
Visitor	Occasional user who opens the main website without signing in.	To find a suitable room and see when it is free before deciding to register or book.	• Browse rooms and facilities (FR-03) • View room availability (FR-04) • Register an account (FR-01)
Member	Primary user: an employee with a self-registered or administrator-created account (role user).	To reserve a room in advance and keep track of own reservations.	• Sign in / sign out (FR-02) • Create member reservation (FR-05) • View own reservations and history (FR-08) • Cancel own reservation (FR-09) • View notifications (FR-13) • Change password (FR-14)
Walk-in booker	A person at the room or a shared kiosk who has no account or does not want to sign in. Gets no member or administrator access.	To book a free room immediately with minimal steps.	• Browse rooms and availability on the kiosk (FR-03, FR-04) • Create walk-in reservation (FR-06)
Administrator	Authorized operator (role admin). Most privileged and least frequent user class.	To keep room information correct, control access and resolve reservation problems.	• All member functions • Manage rooms (FR-10) • Manage users and roles (FR-11) • View and manage all bookings (FR-12)

3. Software Specific Requirements

3.1 Overall Functional Requirements

FR-No.	Function name	User class	Priority	Dependency
FR-01	Account registration	Visitor	High	None
FR-02	Sign in and sign out	Member, Administrator	High	FR-01
FR-03	Browse rooms and facilities	Visitor, Member, Walk-in booker, Administrator	High	None
FR-04	View room availability	Visitor, Member, Walk-in booker, Administrator	High	FR-03
FR-05	Create member reservation	Member, Administrator	High	FR-02, FR-04, FR-07
FR-06	Create walk-in reservation	Walk-in booker	High	FR-04, FR-07
FR-07	Validate reservation	Member, Walk-in booker, Administrator (performed by the system)	High	FR-03
FR-08	View my reservations and history	Member, Administrator	Medium	FR-02, FR-05
FR-09	Cancel reservation	Member (owner), Administrator	High	FR-08 or FR-12
FR-10	Manage room information	Administrator	High	FR-02
FR-11	Manage users and roles	Administrator	Medium	FR-02
FR-12	View and manage all bookings	Administrator	High	FR-02, FR-05, FR-06
FR-13	Record events and show notifications	Member, Administrator	Medium	FR-05, FR-06, FR-09
FR-14	Change password	Member, Administrator	Low	FR-02

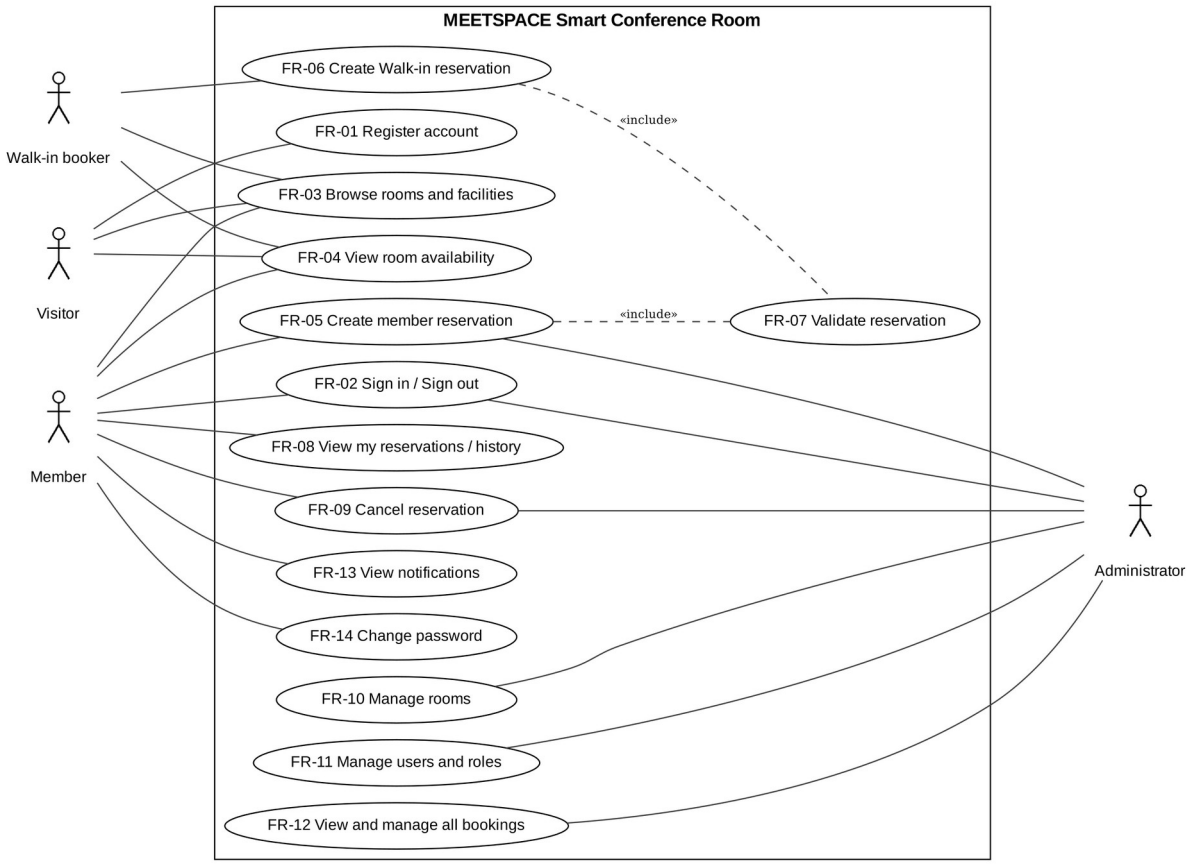


Figure 3 Use case diagram

3.2 Functional Requirements

FR-01	**Account registration**
Description	The system shall allow a visitor to create a member account (role user) and shall sign the new member in after the account is created.
User Class (Actors)	Visitor
Priority	High
Input	• Name: text, 1-100 characters, required • Email: valid email format (name@domain.tld), max 160 characters, unique (case-insensitive), required • Password: 8-12 characters, required • Confirm password: must equal Password, required (checked in the browser)
Output	• Success: account created, user signed in, message "Your account is ready. Welcome to MeetSpace." and the Dashboard or the page the user originally requested is shown. • Error: "Please check the required fields." (invalid name or email), "An account with this email already exists.", "Use a password between 8 and 12 characters.", "The passwords do not match."
Workflow	1. The visitor selects Sign in and then Create an account.

	2. The visitor enters name, email, password and confirm password and submits. 3. The system validates the input. a. If any field is invalid, the system shows the related error message and creates no account. b. If the email already exists, the system shows "An account with this email already exists." 4. The system stores the password as a BCrypt hash, saves the account with role user, and starts a session. 5. The system shows the success message and opens the Dashboard (or the requested page).

FR-02	**Sign in and sign out**
Description	The system shall verify the email and password of an account, start a session with the role of that account, and end the session when the user signs out.
User Class (Actors)	Member, Administrator
Priority	High
Input	• Email: text, required • Password: 8-12 characters, required • Remember me: checkbox, optional (sent to the server but not used in Release 1; every session expires after 30 days of inactivity) • Sign out: button
Output	• Success: the user is signed in and returned to the requested page or the Dashboard; the menu shows Administration only for an administrator. • Error: "The email or password is incorrect." (HTTP 401); no session is created. • Sign out: session invalidated and the guest view is shown.
Workflow	1. The user selects Sign in and enters email and password. 2. The system finds the account by email (case-insensitive) and compares the password with the stored BCrypt hash. 3. The system checks the result. a. If correct, the system creates a session and shows the requested page or the Dashboard. b. If incorrect, the system shows "The email or password is incorrect." 4. When the user selects Sign out, the system invalidates the session and shows the guest Dashboard.

FR-03	**Browse rooms and facilities**
Description	The system shall display the list of meeting rooms that are not deleted, with photo, room code, name (English or Thai), location, capacity, facilities, status and a daily timeline, and shall allow the user to search and choose a date.
User Class (Actors)	Visitor, Member, Walk-in booker, Administrator
Priority	High
Input	• Search text: free text matched against room name, location and room code, optional • Date: date picker, default today, optional • Language: English or Thai (Settings)

Output	• Room cards showing room code, photo, name, location, capacity, facility icons (Display, Wi-Fi, Whiteboard, Projector, Video conferencing, Audio, Computers), status badge (Available, Reserved, Unavailable, Outside opening hours) and the 08:00-20:00 timeline. • If no room matches: "No rooms" empty state with a Clear filters button.
Workflow	1. The user opens Meeting rooms (or the walk-in kiosk). 2. The system loads all rooms and hides rooms marked deleted. 3. The user types search text and/or selects a date. 4. The system filters the cards and recalculates each room's status for the selected date. 5. The user selects Book on a room to start FR-05 or FR-06.

FR-04	**View room availability**
Description	The system shall show the occupied time slots of each room for the selected date from active reservations, without exposing another person's name, email, title or walk-in booker identity to visitors, walk-in users or ordinary members.
User Class (Actors)	Visitor, Member, Walk-in booker, Administrator
Priority	High
Input	• Room and date selected in the room list or booking form
Output	• Timeline 08:00-20:00 in 30-minute blocks: free blocks and occupied blocks (Pending or Confirmed reservations). • Visitor / kiosk data: only room, date, start, end and status of active reservations. • Member data: full details of own reservations; anonymous slots for others. • Administrator data: full details including booker and channel.
Workflow	1. The browser requests /api/data (main website) or /api/walk-in/data (kiosk). 2. The system determines the caller (none, member or administrator); the kiosk endpoint is always treated as anonymous. 3. The system returns reservations filtered according to the caller. 4. The browser draws the timeline and refreshes the data before a booking is submitted.

FR-05	**Create member reservation**
Description	The system shall allow a signed-in member or administrator to reserve one room for one date and time range, and shall save a valid request immediately as Confirmed.
User Class (Actors)	Member, Administrator
Priority	High
Input	• Room: selected active room, required • Date: YYYY-MM-DD, required • Start time and End time: HH:mm in 30-minute steps between 08:00 and 20:00, required • Request key: generated automatically by the browser (max 80 characters) • No meeting title, participant count or repeat option is requested; the server stores title "Room reservation" and participants = 1.
Output	• Success: reservation saved with status Confirmed, a confirmed event is created and

	the message "Reservation submitted successfully." / "Your reservation is saved. You can track it in My reservations." is shown. • Error: the FR-07 error messages; if not signed in, the sign-in screen is shown and the user returns to the booking after signing in.
Workflow	1. The member selects Book on a room (a visitor is asked to sign in first). 2. The member chooses date, start time and end time; the form shows the timeline of the selected day. 3. The member selects Review, checks the summary and selects Confirm. 4. The system validates the request (FR-07). 5. The system checks the result. a. If valid, the system saves the reservation as Confirmed and records a confirmed event. b. If invalid, the system shows the error message and saves nothing; the user's entries are kept.

FR-06	**Create walk-in reservation**
Description	The system shall allow a person without an account to reserve a room from the separate kiosk page /#walkin by declaring a name or employee ID, and shall save a valid request as Confirmed under the system walk-in account.
User Class (Actors)	Walk-in booker
Priority	High
Input	• Room: selected active room, required • Booker name or employee ID: text, 1-100 characters, required • Date, Start time, End time: same rules as FR-05, required • Request key: generated automatically
Output	• Success: reservation saved as Confirmed with the declared booker; the kiosk shows "Your room is booked. Save this reference before closing." and the booking reference. • Error: "Enter a name or employee ID (max 100 characters)." or the FR-07 error messages. • The kiosk does not sign the user in and never lists earlier walk-in bookers.
Workflow	1. The user opens /#walkin (no sidebar link exists on the main website). 2. The user selects Book on a room and enters a name or employee ID, date, start and end time. 3. The user reviews and confirms. 4. The system validates the request (FR-07) and the booker field. 5. The system checks the result. a. If valid, the system saves the reservation under the walk-in account, stores the booker, records an event and shows the reference. b. If invalid, the system shows the error message and saves nothing.

FR-07	**Validate reservation**
Description	The system shall validate every reservation request on the server before saving it, for both the member and the walk-in channel.
User Class (Actors)	Member, Walk-in booker, Administrator (performed by the system)

Priority	High
Input	• Room ID, date, start time, end time, request key and (walk-in only) booker
Output	• Valid: the request continues to FR-05 / FR-06. • Errors: "This item could not be found." (room missing or deleted); "Unavailable" (room inactive); "Choose a valid time between 08:00 and 20:00, in 30-minute intervals. The end must be after the start."; "Choose a valid date."; "Reservations must be in the future."; "Reservations can be made up to one year ahead."; "This time overlaps another reservation. Please choose a different time."; "This submission has already been used. Please start a new reservation."
Workflow	1. Check that the room exists, is not deleted and is active. 2. Check that start and end are on 30-minute boundaries, start < end, start >= 08:00, end <= 20:00 and the duration is at most 4 hours. 3. Check that the date is valid, not later than one year from today (Asia/Bangkok), and that the start time is later than the current time when the date is today. 4. Check that the same account has not used the request key before. 5. Check that no Pending or Confirmed reservation of the same room and date overlaps the interval (existing start < new end and existing end > new start). 6. If any check fails, return the matching error and save nothing.

FR-08	**View my reservations and history**
Description	The system shall show the signed-in user's own reservations split into My reservations (active and not yet ended) and Reservation history (cancelled or ended).
User Class (Actors)	Member, Administrator
Priority	Medium
Input	• Signed-in session
Output	• Table with room name, date, start-end time, status badge (Confirmed in green / Cancelled in red), Details and Cancel buttons. • Details dialog: status, room, date, time, requester, reference, created and last updated time. • Empty state when there are no reservations.
Workflow	1. The user opens My reservations or Reservation history. 2. The system returns the user's own reservations. 3. The browser places active future reservations in My reservations and cancelled or ended ones in Reservation history. 4. The user may open Details or start a cancellation (FR-09).

FR-09	**Cancel reservation**
Description	The system shall allow the owner of a reservation or an administrator to cancel an active reservation that has not yet ended; the reservation becomes Cancelled and its time slot is released.
User Class (Actors)	Member (owner), Administrator
Priority	High
Input	• Reservation ID (selected from the list) • Confirmation in the cancel dialog
Output	• Success: status Cancelled, updated time recorded, cancelled event created, message

	"Reservation cancelled." • Error: "You don't have permission to perform this action." (not owner and not admin); "This reservation has changed. Refresh and try again." (already cancelled); "This reservation has already ended and cannot be cancelled."
Workflow	1. The user selects Cancel on a reservation. 2. The system shows "The room will become available for other people. This action cannot be undone." 3. The user confirms. 4. The system checks permission and state. a. If the user is not the owner or an administrator, the system rejects the request. b. If the reservation is not active or has ended, the system rejects the request. 5. The system sets the status to Cancelled, records a cancelled event and refreshes the lists. 6. A walk-in booker cannot cancel from the kiosk; an administrator cancels walk-in reservations (FR-12).

FR-10	**Manage room information**
Description	The system shall allow an administrator to add, edit and delete meeting rooms, including choosing or uploading a room photo.
User Class (Actors)	Administrator
Priority	High
Input	• Room name (English) and Room name (Thai): 1-100 characters, required • Location (English) and Location (Thai): 1-120 characters, required • Capacity: whole number 1-1000, required • Colour: indigo, teal, blue or violet • Facilities: any of Display, Wi-Fi, Whiteboard, Projector, Video conferencing, Audio, Computers • Active: yes/no • Photo: a bundled photo or an uploaded PNG/JPG up to 3 MB, at most 6000 px per side and 16 megapixels • Room ID: generated automatically (R001, R002, ...) for a new room
Output	• Success: "Meeting room saved successfully." or "Meeting room deleted." • Error: "Please check the required fields."; "Choose a valid PNG or JPG, up to 3 MB and 16 megapixels."; "This change affects upcoming reservations. Cancel those reservations first or keep sufficient capacity." (deactivating a room with upcoming reservations); "This room still has upcoming reservations. Cancel them before deleting the room."
Workflow	1. The administrator opens Administration > Room management. 2. The administrator selects Add room or Edit, fills in the form and saves. 3. The system validates the fields and the photo, and rejects deactivation when the room has upcoming active reservations. 4. The system saves the room and refreshes the list. 5. To delete, the administrator selects Delete and confirms; if the room has no upcoming active reservation the system marks it deleted and inactive (soft delete), hides it from room lists and booking, and keeps past reservation history.

FR-11	**Manage users and roles**
Description	The system shall allow an administrator to add team member accounts and to change the role of other accounts between Member and Administrator.
User Class (Actors)	Administrator
Priority	Medium
Input	• Add member: Name (1-100 characters), Email (valid, unique), Password (8-12 characters), Role (Member or Administrator), all required • Change role: target account, new role, previous role (sent automatically to detect concurrent changes)
Output	• Success: "Team member added. Share their credentials securely." / "Account permissions updated." • Error: "You cannot change your own permissions or the workspace owner's permissions."; "These permissions have changed. Refresh and try again."; "An account with this email already exists."; password length and required-field messages.
Workflow	1. The administrator opens Administration > Members and roles. 2. To add, the administrator fills in the form; the system validates and creates the account with a BCrypt password hash. 3. To change a role, the administrator selects Manage on an account and chooses the new role. 4. The system checks the target. a. If the target is the administrator's own account or the seeded workspace administrator, the change is rejected. b. If the previous role no longer matches, the change is rejected. 5. The system saves the change and refreshes the list. Accounts cannot be deleted in Release 1.

FR-12	**View and manage all bookings**
Description	The system shall show administrators all reservations in two separate lists, Walk-in bookings and Member bookings, with booker, room, date, time and status, filterable by reservation date and room, and shall allow cancellation.
User Class (Actors)	Administrator
Priority	High
Input	• Reservation date: calendar date, optional • Room: any room including deleted rooms (labelled), optional • Selected reservation for Details or Cancel
Output	• Two panels with the number of bookings in each; rows sorted by start time when a date is selected, otherwise by newest creation time. • Empty state "No bookings found" when nothing matches.
Workflow	1. The administrator opens Administration > Booking management. 2. The administrator chooses a date and/or room (Clear and Refresh buttons are available). 3. The system shows walk-in and member reservations in separate lists with booker name and channel. 4. The administrator opens Details or cancels an active future reservation (FR-09).

FR-13	Record events and show notifications
Description	The system shall record an event each time a reservation is confirmed or cancelled, and shall show the signed-in user the events of their own reservations, newest first.
User Class (Actors)	Member, Administrator
Priority	Medium
Input	• Reservation ID, recipient account, event type (confirmed or cancelled), event time and actor name - created automatically by FR-05, FR-06 and FR-09
Output	• Notifications page listing "Reservation confirmed" or "Reservation cancelled" with room, date and time; the bell icon opens it. • Note shown: "Updates appear in MeetSpace. Email and Google Calendar are not connected."
Workflow	1. When a reservation is confirmed or cancelled, the system stores an event for the reservation owner. 2. When the user opens Notifications, the system returns the user's events ordered by event time descending. 3. Events of walk-in reservations are stored for the walk-in system account and are not shown to any person.

FR-14	Change password
Description	The system shall allow a signed-in user to change their password after verifying the current password, and shall end the session so the user signs in again.
User Class (Actors)	Member, Administrator
Priority	Low
Input	• Current password: required • New password: 8-12 characters, required
Output	• Success: "Password changed. Please sign in again." and the session ends. • Error: "The email or password is incorrect." (wrong current password); "Use a password between 8 and 12 characters."; "Please sign in to continue." (no session).
Workflow	1. The user opens Settings > Account security. 2. The user enters the current and new password and submits. 3. The system verifies the current password and the new password length. a. If a check fails, the system shows the error and keeps the old password. 4. The system stores the new BCrypt hash, invalidates the session and asks the user to sign in again.

3.3 Non-Functional Requirements

NFR-No.	Name	Requirement Statement	Acceptance Criteria	Verification Method	Related FR
NFR-01	Security - password storage	The system shall store every password as a BCrypt hash and shall not return password hashes in any API response.	No readable password exists in APP_USER.password_hash; /api/session, /api/data and login responses contain no password field.	Inspection of database and API responses	FR-01, FR-02, FR-11, FR-14

NFR-No.	Name	Requirement Statement	Acceptance Criteria	Verification Method	Related FR
NFR-02	Security - access control	The system shall return HTTP 401 for protected operations without a session and HTTP 403 when a non-administrator calls an administrator operation.	Calls to /api/book, /api/reservation and /api/password without a session return 401; a member calling /api/room, /api/room-delete, /api/user or /api/user-role receives 403.	Automated API test (ApiControllerTest)	FR-05, FR-09, FR-10, FR-11, FR-12
NFR-03	Privacy	The system shall not include account names, email addresses, reservation titles or walk-in booker identities of other people in visitor, kiosk or ordinary-member availability data.	Responses of /api/data (no session or member session) and /api/walk-in/data contain only room, date, start, end and status for other people's reservations.	Automated API test and inspection	FR-04, FR-06
NFR-04	Reliability - no double booking	The system shall not allow two active reservations of the same room to overlap, including when a member and a walk-in request arrive at the same time.	In a test that submits 10 simultaneous overlapping requests through both channels, exactly one reservation is saved and the others receive the conflict error.	Concurrency test	FR-05, FR-06, FR-07
NFR-05	Correctness - time zone	The system shall evaluate past time, today, one-year limit and reservation end using the Asia/Bangkok time zone.	A request for today at a start time already passed in Bangkok is rejected; a request at the next 30-minute slot is accepted, regardless of the server's own time zone.	Boundary test	FR-07, FR-09
NFR-06	Data integrity	A rejected reservation request shall leave no reservation and no event record.	After 20 invalid requests (each validation rule), the number of rows in BOOKING and BOOKING_EVENT is unchanged.	Automated test with row counts	FR-07, FR-13
NFR-07	Persistence	The system shall keep accounts, rooms, reservations, events and uploaded photos after the server is restarted.	Data created before a restart is shown unchanged after the restart.	Restart test	FR-01, FR-05, FR-06, FR-10, FR-13
NFR-08	Performance (target)	The system should complete 95% of /api/data and /api/book requests within 2 seconds with 20 concurrent users and 1,000 stored reservations.	95th percentile response time <= 2 s in the documented test environment.	Performance test (planned; not yet executed)	FR-03, FR-04, FR-05
NFR-09	Compatibility / responsive layout	Every screen shall be usable at 1366 x 768 (desktop) and 390 x 844 (mobile) viewports in a current Chrome, Edge or Safari browser.	No control is clipped or overlapped and no horizontal page scroll appears at either size.	Manual inspection at both viewports	FR-03, FR-05, FR-06
NFR-10	Usability - status display	The system shall show each reservation status as a text label with colour: Confirmed in green and Cancelled in red, so that colour is not the only cue.	Every status badge shows the words Confirmed / Cancelled (or Thai equivalents) with the stated colour.	UI test (ui.test.cjs) and inspection	FR-08, FR-12
NFR-11	Localization	The system shall provide the complete interface in English and Thai and shall remember	Switching language changes all labels, messages and room names; the choice is	Inspection	FR-03, FR-05, FR-06

NFR-No.	Name	Requirement Statement	Acceptance Criteria	Verification Method	Related FR
		the selected language and theme in the browser.	kept after reloading the page.		
NFR-12	Security - uploaded files	The system shall accept only PNG or JPEG room photos of at most 3 MB, 6000 px per side and 16 megapixels, and shall reject external URLs and SVG files.	Valid PNG/JPG files are saved; an SVG, a 4 MB file or a URL is rejected with the invalid image message.	Automated API test	FR-10
NFR-13	Security - session	A signed-in session shall end after 30 days of inactivity, when the user signs out, or after a password change.	After sign-out or password change, a protected request with the old session returns 401.	Test and configuration inspection	FR-02, FR-14

4. User Interface Requirements

4.1 Navigation Structure / Sitemap

Main website: Dashboard, Meeting rooms (booking form, review, result), My reservations (details, cancel), Reservation history, Notifications and Settings (language, theme, change password, sign out); Sign in / Register; Administration (administrators only) with Booking management, Room management and Members and roles. The walk-in kiosk is a separate entry at /#walkin (room list, booking form with booker name or employee ID, review, result with reference) and has no link in the main sidebar. In Figure 4, one asterisk marks pages that require sign-in and two asterisks mark pages for administrators only.

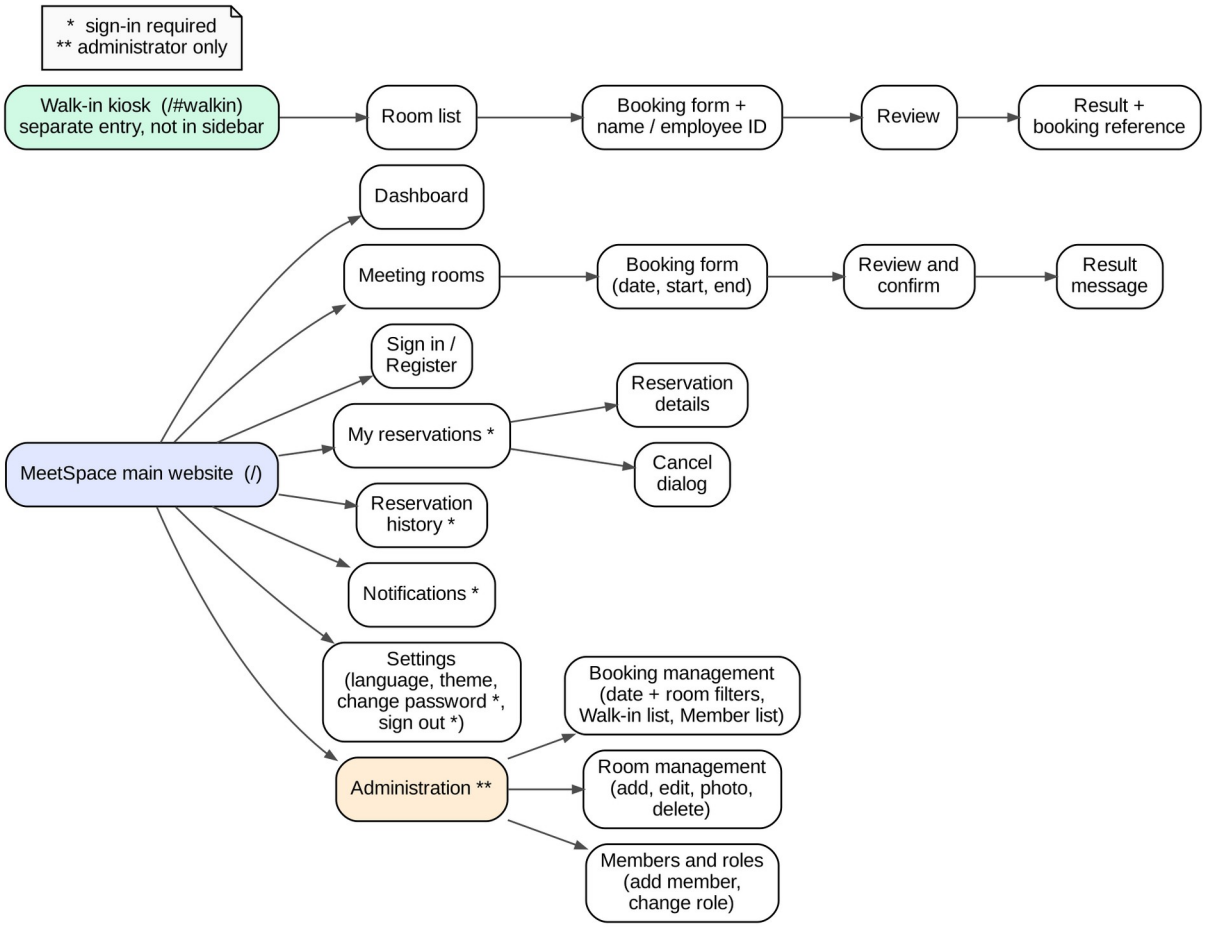
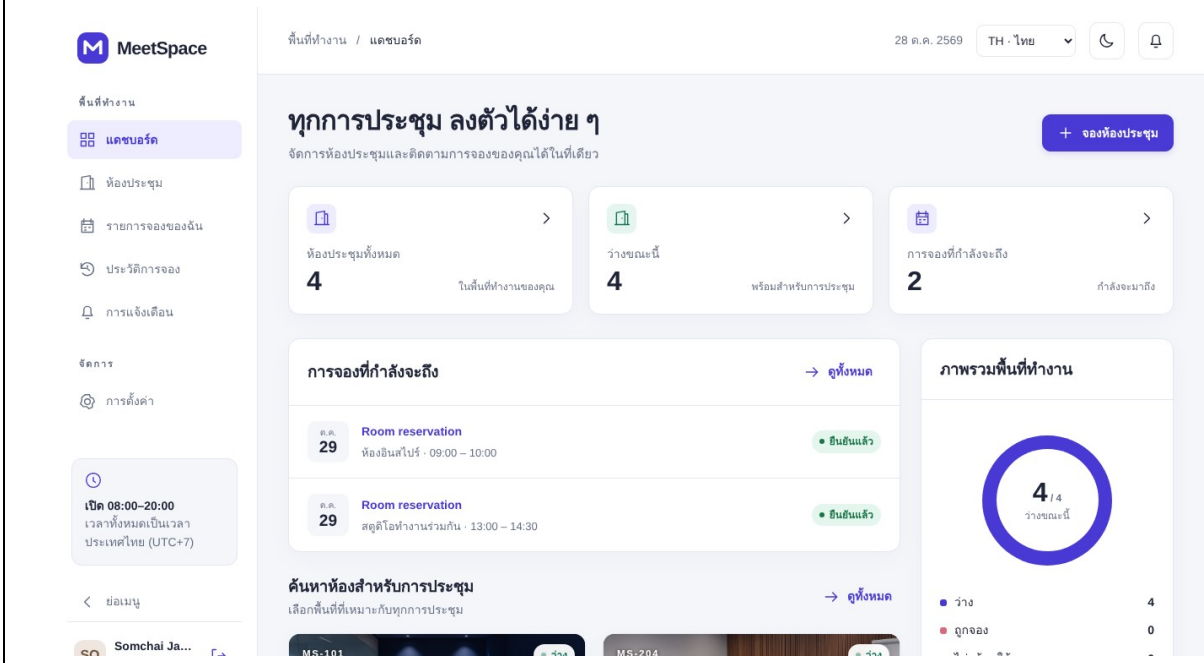
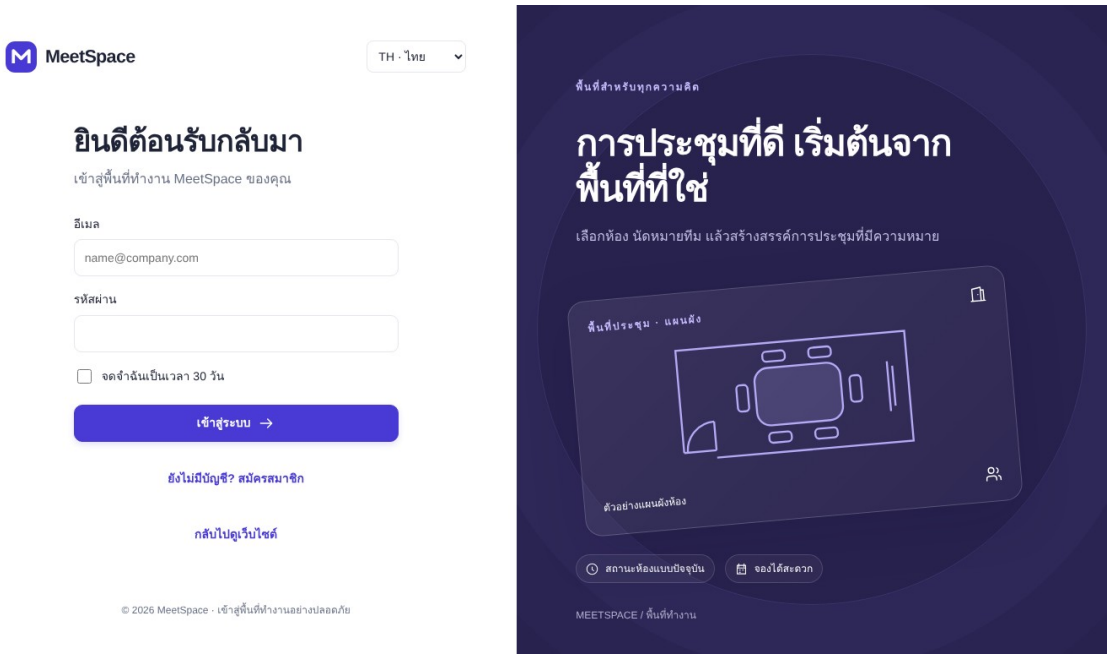


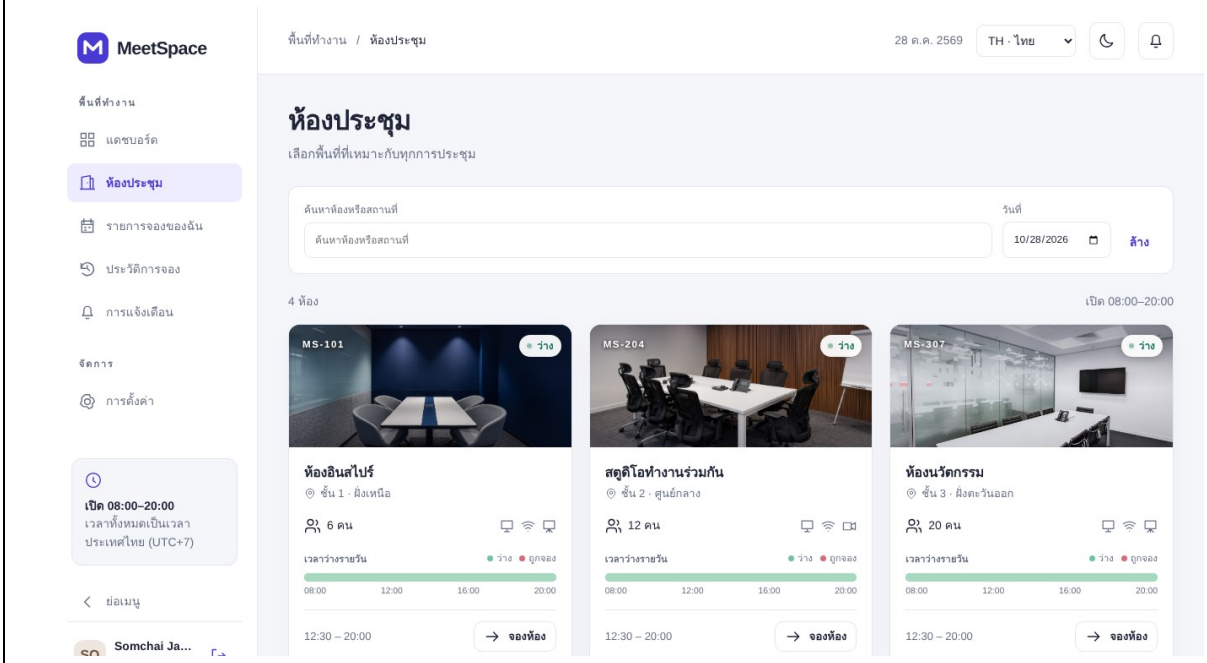
Figure 4 Navigation structure / sitemap

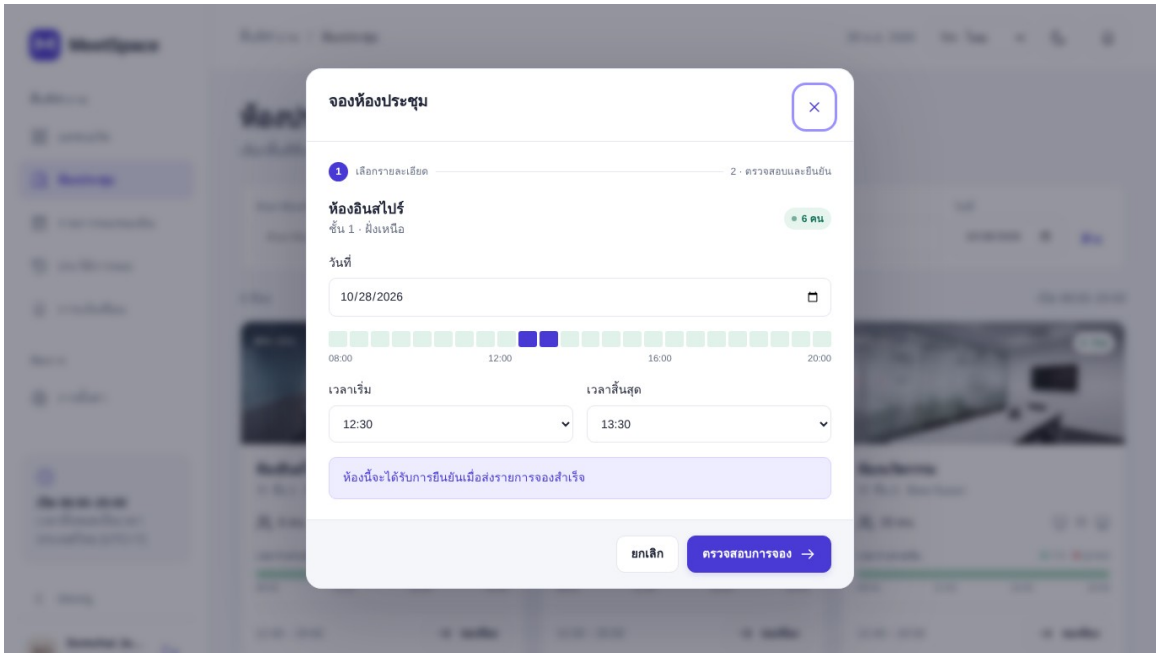
4.2 User Interface Specifications

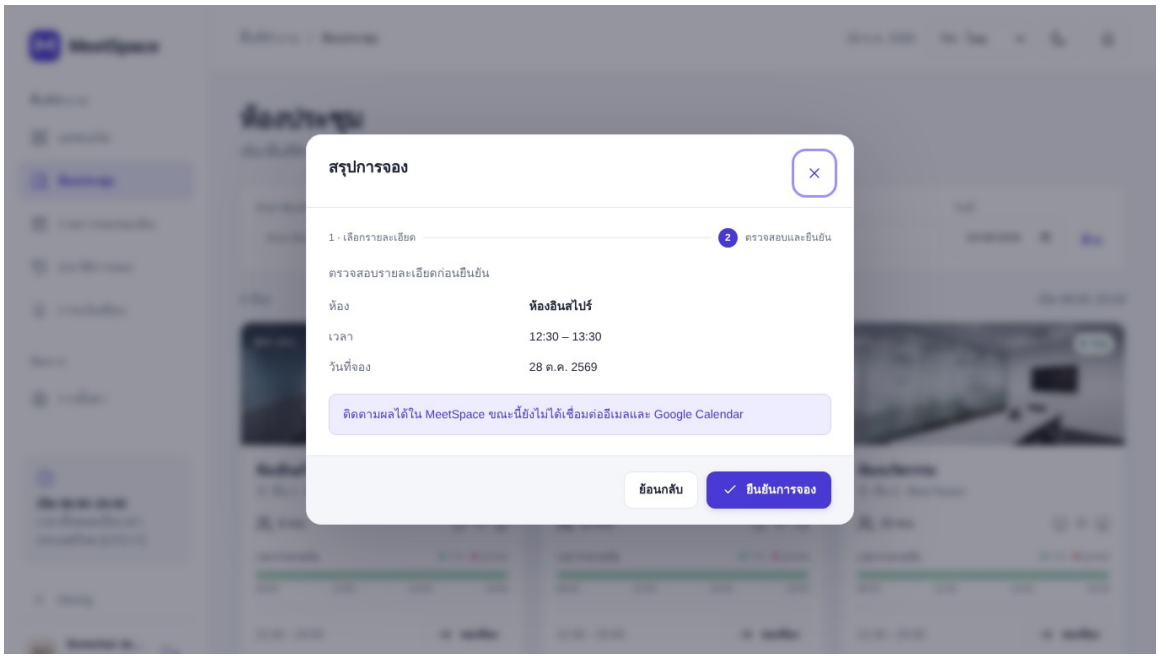
Screens were captured from the current application (Thai interface, 1366 x 768). Visual design may evolve; the required information and behaviour are stated in each table.

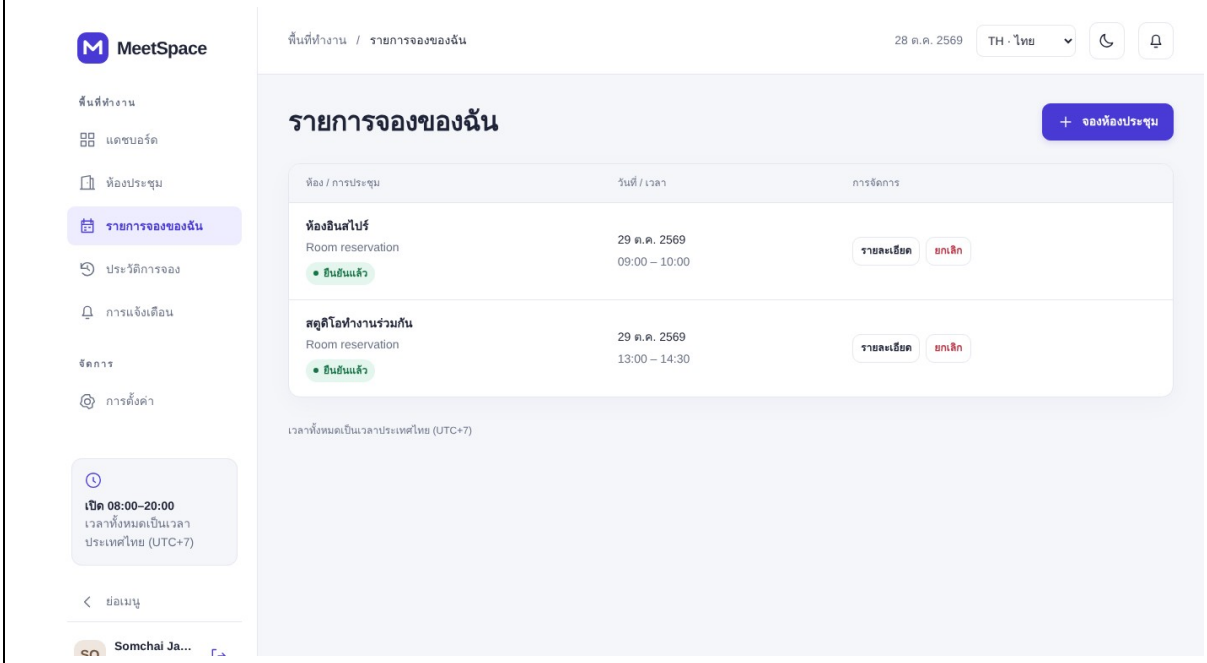
UI-01:	**Dashboard**
	
Purpose:	Give an overview of total rooms, rooms available now, the user's upcoming reservations, workspace occupancy and a quick link to book.
Prerequisite:	1. The user opens the main website (/). Visitors see the same page without personal data.
Use case procedure:	1. The user reads the three summary cards. 2. The user opens an upcoming reservation or selects Book a room.
Acceptance Criteria	1. Counts match the current data. 2. Upcoming reservations list only the signed-in user's active future reservations. 3. Visitors see no personal reservation data.
Related Function Requirement	FR-03, FR-04, FR-08

UI-02:	**Sign in and registration**
	
Purpose:	Allow a visitor to sign in or create an account.
Prerequisite:	1. The visitor selects Sign in, or tries to open a page that requires sign-in.
Use case procedure:	1. The user enters email and password (and Remember me) and selects Sign in. 2. A new user selects Create an account and enters name, email, password and confirm password.
Acceptance Criteria	1. Correct credentials open the requested page or the Dashboard. 2. Wrong credentials show "The email or password is incorrect." 3. Passwords outside 8-12 characters are rejected.
Related Function Requirement	FR-01, FR-02

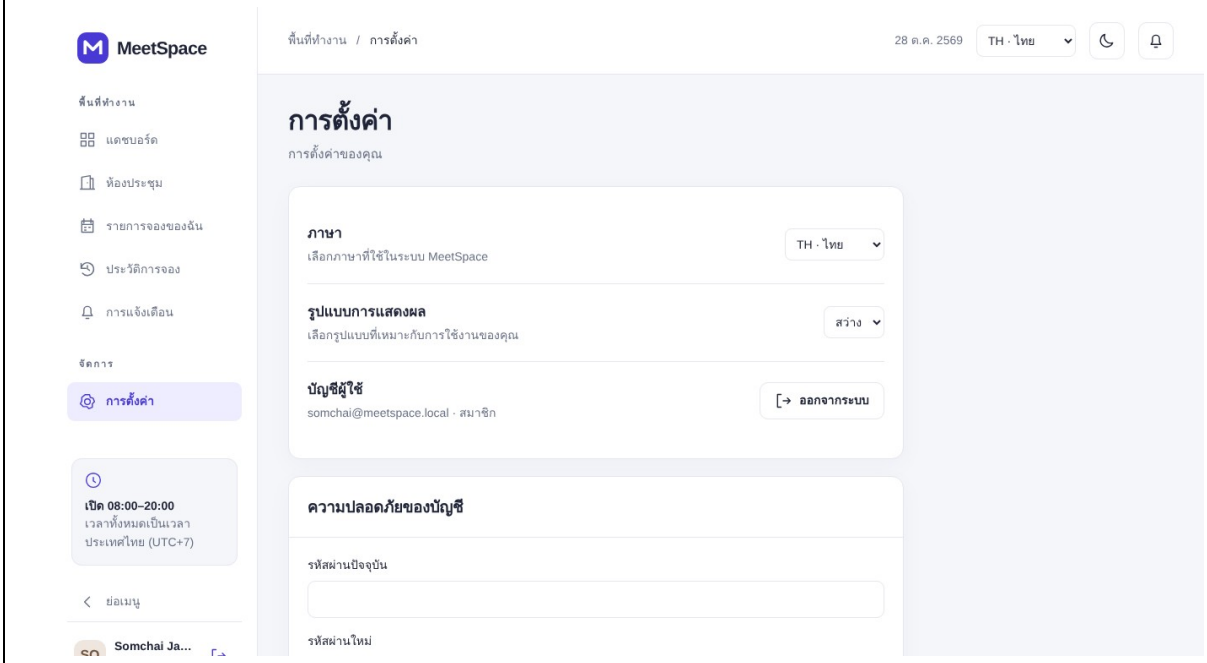
UI-03:	**Meeting rooms**
	
Purpose:	Search rooms and inspect photo, location, capacity, facilities, status and the daily timeline.
Prerequisite:	1. Select Meeting rooms in the sidebar.
Use case procedure:	1. The user types search text or selects a date. 2. The user reads the timeline (green = free, red = occupied). 3. The user selects Book on a room.
Acceptance Criteria	1. Only rooms that are not deleted are shown. 2. Search matches name, location or room code. 3. Occupied slots show no identity of other users.
Related Function Requirement	FR-03, FR-04

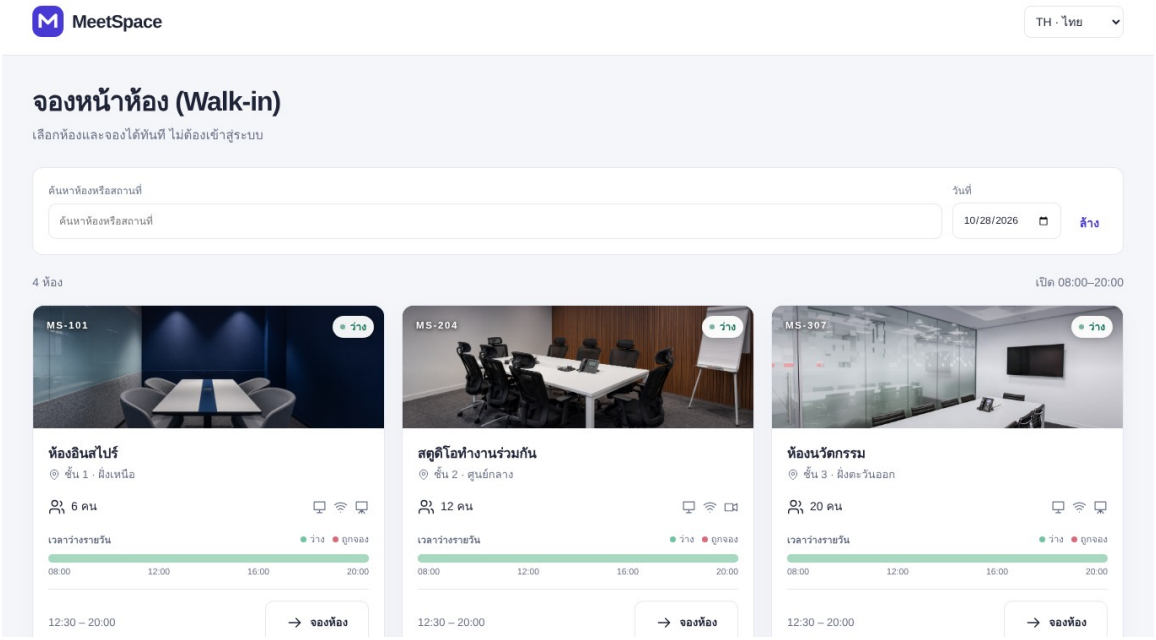
UI-04:	**Booking form (member)**
	
Purpose:	Choose date, start time and end time for the selected room.
Prerequisite:	1. A signed-in user selects Book on a room (a visitor is asked to sign in first).
Use case procedure:	1. The user selects a date; the timeline for that date is shown. 2. The user selects start and end time (30-minute steps). 3. The user selects Review booking.
Acceptance Criteria	1. Times outside 08:00-20:00, longer than 4 hours or with end <= start cannot be submitted. 2. No meeting title, participant or repeat field is shown. 3. The note says the room is confirmed immediately.
Related Function Requirement	FR-05, FR-07

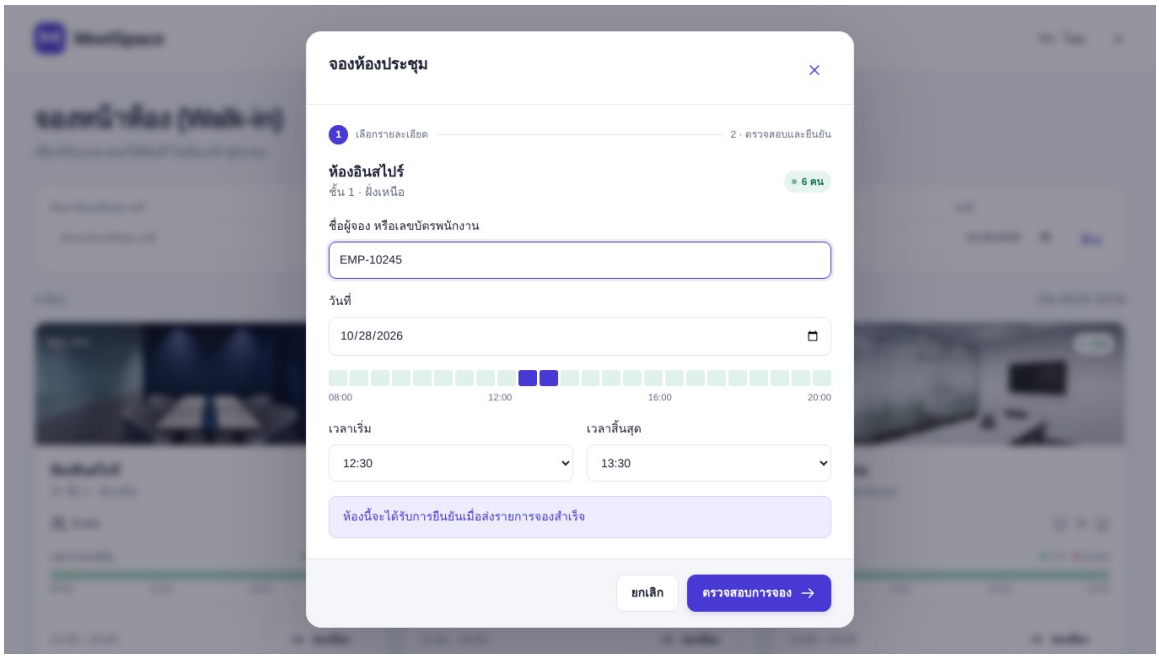
UI-05:	**Booking review**
	
Purpose:	Show the summary of the request before it is submitted.
Prerequisite:	1. The user completed UI-04.
Use case procedure:	1. The user checks room, time and date. 2. The user selects Confirm (or Back to edit).
Acceptance Criteria	1. A valid request is saved as Confirmed and a success message is shown. 2. A conflict or validation error is shown in the dialog and nothing is saved.
Related Function Requirement	FR-05, FR-07

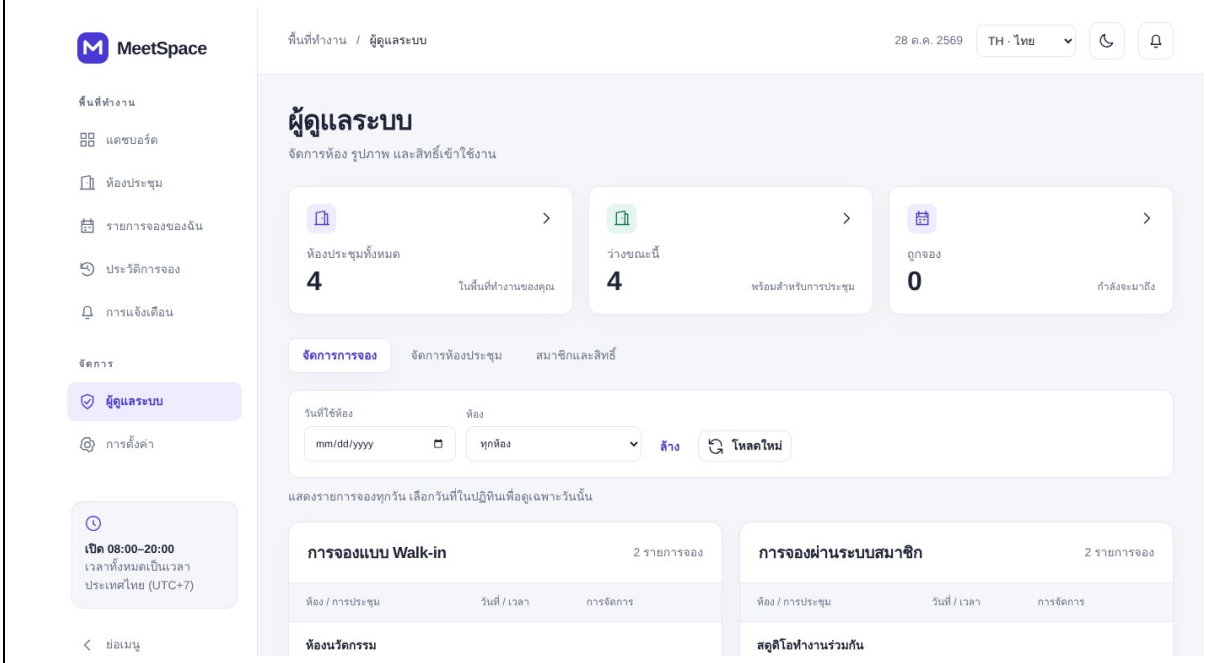
UI-06:	**My reservations and history**
	
Purpose:	List the user's own reservations with status and actions.
Prerequisite:	1. Signed-in user selects My reservations or Reservation history.
Use case procedure:	1. The user views the list. 2. The user selects Details, or Cancel and confirms.
Acceptance Criteria	1. Active future reservations appear in My reservations; cancelled or ended ones in Reservation history. 2. Status shows Confirmed (green) or Cancelled (red). 3. Cancel is shown only for active reservations that have not ended.
Related Function Requirement	FR-08, FR-09

UI-07:	**Notifications**
	
Purpose:	Show confirmation and cancellation events of the user's reservations.
Prerequisite:	1. Signed-in user selects Notifications or the bell icon.
Use case procedure:	1. The user reads the list and may open the related reservation.
Acceptance Criteria	1. Events are ordered newest first. 2. Each event shows type, room, date and time. 3. The page states that email and Google Calendar are not connected.
Related Function Requirement	FR-13

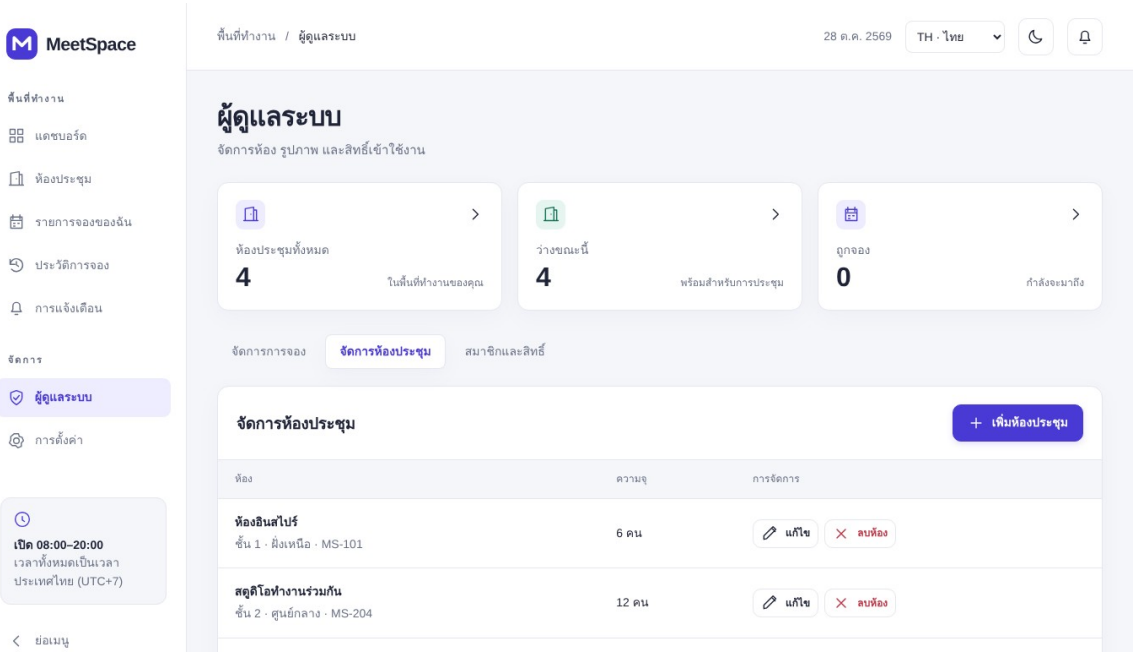
UI-08:	**Settings**
	
Purpose:	Change language and theme, sign out and change password.
Prerequisite:	1. Select Settings in the sidebar (account section requires sign-in).
Use case procedure:	1. The user changes language or theme. 2. The user enters current and new password and saves. 3. The user selects Sign out.
Acceptance Criteria	1. Language and theme apply immediately and are remembered. 2. A successful password change ends the session and shows "Password changed. Please sign in again."
Related Function Requirement	FR-02, FR-14

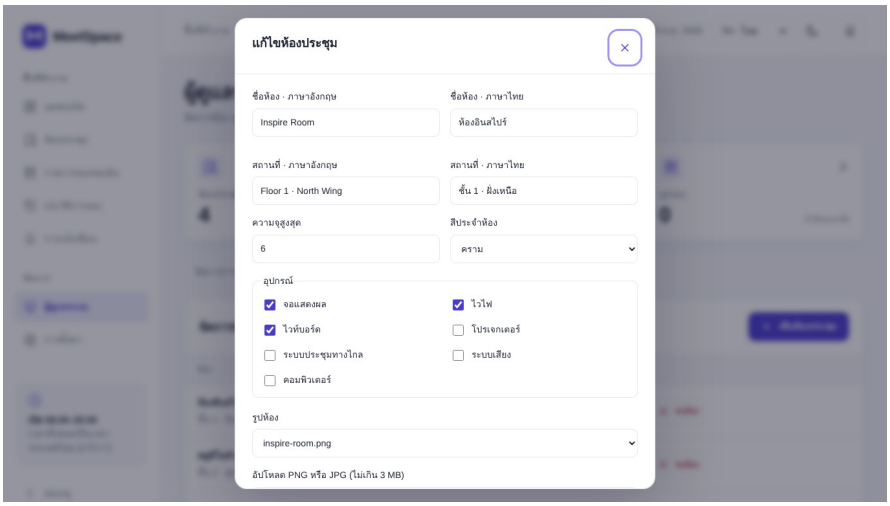
UI-09:	**Walk-in kiosk - room list**
	
Purpose:	Let a person without an account choose a room on the separate kiosk page.
Prerequisite:	1. Open /#walkin on a shared device. No sign-in is needed and the main sidebar is not shown.
Use case procedure:	1. The user searches or selects a date. 2. The user selects Book on a free room.
Acceptance Criteria	1. The page works without a session and never shows earlier bookers. 2. Occupied slots are shown anonymously.
Related Function Requirement	FR-03, FR-04, FR-06

UI-10:	**Walk-in kiosk - booking form**
	
Purpose:	Enter booker name or employee ID and choose a time.
Prerequisite:	1. The user selected a room in UI-09.
Use case procedure:	1. The user enters a name or employee ID (required). 2. The user selects date, start and end time. 3. The user reviews and confirms.
Acceptance Criteria	1. An empty name or more than 100 characters is rejected. 2. A valid request is confirmed immediately and the booking reference is displayed. 3. The user is not signed in after booking.
Related Function Requirement	FR-06, FR-07

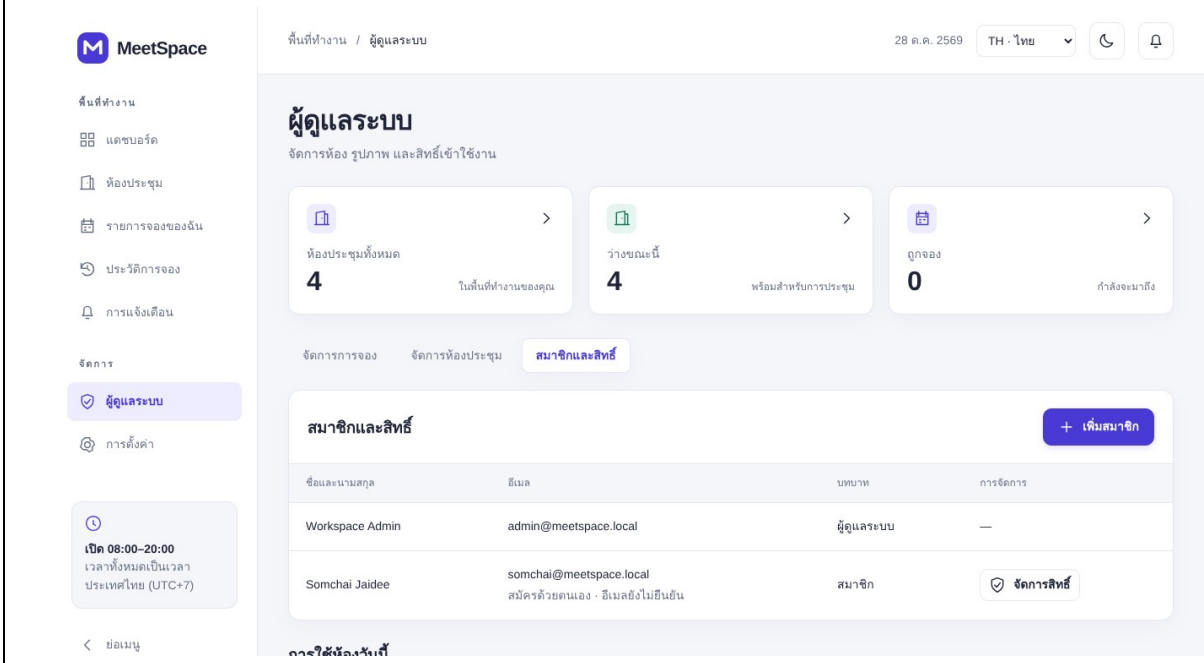
UI-11:	**Administration - Booking management**
	
Purpose:	Let an administrator review and cancel all reservations.
Prerequisite:	1. Administrator signs in and selects Administration > Booking management.
Use case procedure:	1. The administrator selects a reservation date and/or room. 2. The administrator reviews the Walk-in bookings and Member bookings lists. 3. The administrator opens Details or cancels a reservation.
Acceptance Criteria	1. Walk-in and member reservations are shown in separate lists with booker and channel. 2. Filters by date and room work together; Clear resets them. 3. Only administrators can open this page.
Related Function Requirement	FR-12, FR-09

UI-12:	**Administration - Room management**





Purpose:	Let an administrator add, edit, change the photo of and delete rooms.
Prerequisite:	1. Administrator selects Administration > Room management.
Use case procedure:	1. The administrator selects Add room or Edit and fills in the room form (figure below). 2. The administrator selects Delete and confirms the warning.
Acceptance Criteria	1. Invalid fields or photos are rejected with a message. 2. A room with upcoming reservations cannot be deactivated or deleted. 3. A deleted room disappears from room lists but stays in booking history.
Related Function Requirement	FR-10

UI-13:	**Administration - Members and roles**
	
Purpose:	Let an administrator add accounts and change roles.
Prerequisite:	1. Administrator selects Administration > Members and roles.
Use case procedure:	1. The administrator selects Add member and fills in name, email, password and role. 2. The administrator selects Manage on an account to change its role.
Acceptance Criteria	1. New accounts can sign in immediately. 2. The administrator's own account and the workspace administrator are protected. 3. Duplicate emails are rejected.
Related Function Requirement	FR-11

Appendix A: Data Dictionary

The data dictionary follows src/main/resources/schema.sql. Figure 5 shows the entity relationship diagram.

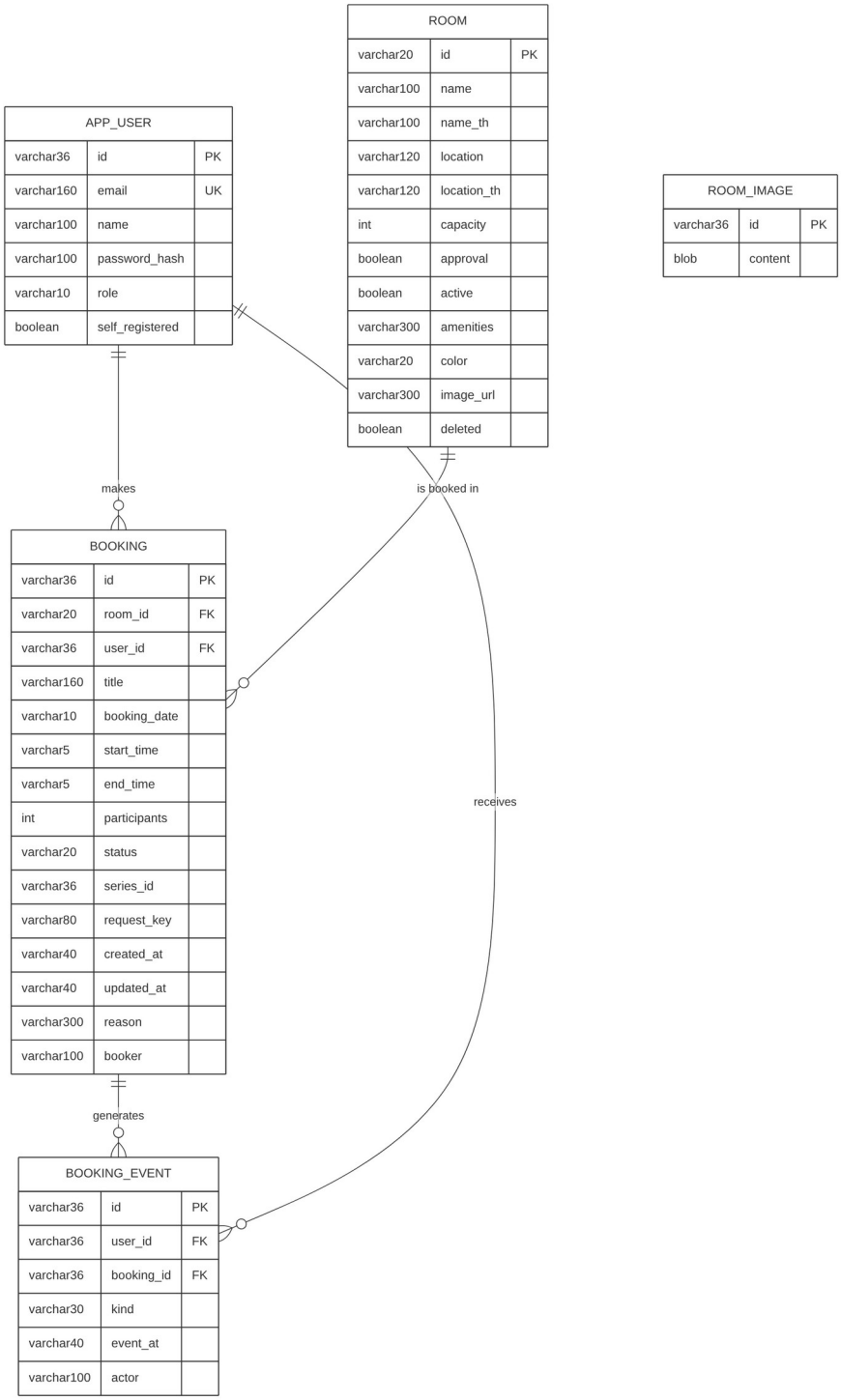


Figure 5 Entity relationship diagram

Table: APP_USER - User accounts, including the seeded workspace administrator and the internal walk-in account.				
Attribute	**Description**	**Data type**	**Key**	**Example**
id	Account identifier (UUID; fixed IDs workspace-admin and walk-in for seeded accounts)	VARCHAR(36)	PK	workspace-admin
email	Sign-in email, unique, stored in lower case	VARCHAR(160)	UK	admin@meetspace.local
name	Display name	VARCHAR(100)		Workspace Admin
password_hash	BCrypt hash of the password	VARCHAR(100)		$2a$10$Qe...k2
role	user or admin	VARCHAR(10)		admin
self_registered	TRUE when created through registration; default FALSE	BOOLEAN		FALSE

Table: ROOM - Meeting rooms. Deleted rooms are kept (soft delete).				
Attribute	**Description**	**Data type**	**Key**	**Example**
id	Room code (letters, digits, - or _, max 20)	VARCHAR(20)	PK	MS-101
name	Room name in English	VARCHAR(100)		Inspire Room
name_th	Room name in Thai	VARCHAR(100)		ห้องอินสไปร์
location	Location in English	VARCHAR(120)		Floor 1 · North Wing
location_th	Location in Thai	VARCHAR(120)		ชั้น 1 · ฝั่งเหนือ
capacity	Maximum number of people (1-1000)	INT		6
approval	Legacy approval flag; always FALSE in Release 1	BOOLEAN		FALSE
active	Room can be booked; default TRUE	BOOLEAN		TRUE
amenities	Comma-separated facility codes; default ''	VARCHAR(300)		display,wifi,whiteboard
color	Interface colour: indigo, teal, blue or violet	VARCHAR(20)		indigo
image_url	Bundled photo path or /api/room-images/{id}	VARCHAR(300)		/images/inspire-room.png
deleted	Soft-delete flag; default FALSE	BOOLEAN		FALSE

Table: BOOKING - Reservations from both channels. UNIQUE (user_id, request_key, booking_date).				
Attribute	**Description**	**Data type**	**Key**	**Example**
id	Reservation identifier (UUID); first 8 characters shown as reference	VARCHAR(36)	PK	3f6c2a91-...-d1e4
room_id	Reserved room	VARCHAR(20)	FK -> ROOM.id	MS-101
user_id	Owner account; walk-in reservations use walk-in	VARCHAR(36)	FK -> APP_USER.id	walk-in

title	Internal label, always "Room reservation"; not entered by users	VARCHAR(160)		Room reservation
booking_date	Date, YYYY-MM-DD	VARCHAR(10)		2026-10-29
start_time	Start, HH:mm	VARCHAR(5)		09:00
end_time	End, HH:mm	VARCHAR(5)		10:00
participants	Compatibility field; always 1	INT		1
status	confirmed or cancelled (pending, rejected are legacy)	VARCHAR(20)		confirmed
series_id	Legacy recurring series ID; NULL in Release 1	VARCHAR(36)		NULL
request_key	Duplicate-submission key	VARCHAR(80)	UK (with user_id, booking_date)	6b1d...-9a0f
created_at	Creation time, ISO-8601 UTC	VARCHAR(40)		2026-10-28T05:16:42Z
updated_at	Last update time, ISO-8601 UTC	VARCHAR(40)		2026-10-28T05:16:42Z
reason	Legacy rejection reason; default ''	VARCHAR(300)		''
booker	Walk-in name/employee ID or member name; default ''	VARCHAR(100)		EMP-10245

Table: BOOKING_EVENT - Confirmation and cancellation events used for notifications.

Attribute	Description	Data type	Key	Example
id	Event identifier (UUID)	VARCHAR(36)	PK	a8e0...-77c2
user_id	Recipient (reservation owner)	VARCHAR(36)	FK -> APP_USER.id	5c1e...-02ab
booking_id	Related reservation	VARCHAR(36)	FK -> BOOKING.id	3f6c...-d1e4
kind	Event type: confirmed or cancelled	VARCHAR(30)		confirmed
event_at	Event time, ISO-8601 UTC	VARCHAR(40)		2026-10-28T05:16:42Z
actor	Name of the account that caused the event	VARCHAR(100)		Somchai Jaidee

Table: ROOM_IMAGE - Uploaded room photos stored as PNG bytes; referenced by ROOM.image_url (no declared FK).

Attribute	Description	Data type	Key	Example
id	Image identifier (UUID)	VARCHAR(36)	PK	c2d4...-8e11
content	PNG image bytes	BLOB		(binary)

Indexes: idx_booking_room_date (room_id, booking_date), idx_booking_user (user_id), idx_event_user (user_id). Relationships: one APP_USER has many BOOKING and many BOOKING_EVENT; one ROOM has many BOOKING; one BOOKING has many BOOKING_EVENT. Dates and times are stored as text in the formats shown.